Estimation of Carbon Storage Based on AAV Multispectral Photogrammetry: An Offshore Islands Study in Pearl River Delta

Zhanyou Mo, Ze Fang, Liren Liu, Lingzhi Chen, Bo Peng, Weiwen Wang, Ming Chang, and Xuemei Wang

***Abstract*—Aboveground biomass (AGB) of forests is a crucial metric for assessing ecosystem carbon storage and cycling. The geographical complexity of offshore islands, especially their perennial cloud cover, poses a challenge to traditional AGB remote sensing, so autonomous aerial vehicle (AAV)-based remote sensing technology is particularly important. However, AAV approaches face limitations from ground interference, structural parameter errors, and allometric equation discrepancies. This study develops a framework integrating AAV photogrammetry and the Mask R-CNN to identify tree species, quantify forest structure, and estimate AGB in a representative offshore island to infer the impact of tree species differences, dominated by allometric equations and structural parameter errors, and dominated by canopy occlusion on the estimated AGB of island forests. Compared to the AGB results calculated by species identification of individual trees, multispectral sensors effectively identified four dominant tree species and land cover, as general allometric equations resulted in significant errors ranging from −68% to +36%. The optimized canopy height model (CHM) approach revealed a 5.6% underestimation of tree height and 10.0% error in diameter at breast height (DBH) due to canopy occlusion. This study provides new insights into AGB estimation and forest species identification methodologies, which are important for studying carbon management activities in similar ecosystem types.**

***Index Terms*—Aboveground biomass (AGB), forest ecosystem, offshore island, tree species identification, autonomous aerial vehicle (AAV).**

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Zhanyou Mo, Ze Fang, Liren Liu, Weiwen Wang, Ming Chang, and Xuemei Wang are with the College of Environment and Climate, Jinan University, Guangzhou 511443, China (e-mail: a806065953@163.com; 15920672736@163.com; 1017236376@qq.com; wwangeci@jnu.edu.cn; changming@email.jnu.edu.cn; eciwxm@jnu.edu.cn).

Lingzhi Chen is with Guangdong Ocean Association, Guangzhou 510220, China (e-mail: soyandreachen@163.com).

Bo Peng is with the Key Laboratory of Marine Environmental Survey Technology and Application, Ministry of Natural Resources, Guangzhou 510300, China (e-mail: pengbo@gig.ac.cn).

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I. Introduction

ABOVEGROUND biomass (AGB) accounts for approximately 50% of ecosystem carbon stocks; thus, biomass carbon sinks contribute significantly to the reduction of greenhouse gas emissions [1]. However, significant uncertainties persist in AGB estimates across different forest types, with a pronounced relative uncertainty range of 9.9%–56.2%, and research is urgently needed to identify the core sources of this estimation bias [2], [3], [4]. Recently, there has been a strong development of remote sensing techniques in AGB estimation for forests, and these new methods can help better understand the uncertainties involved [5], [6].

Islands are challenging ecosystems for carbon stock estimation [7]. On the one hand, ecologists have relied on ground sample measurements and empirical equations as a direct way to obtain accurate AGB at small scales [8]. However, this approach is greatly constrained by the scarce samples of island forests, as well as the prominent spatial and temporal heterogeneity of existing AGB estimates for island ecosystems, which present a relative uncertainty range of 46.5%–122.8% [9], [10], [11], [12]. On the other hand, researchers in the field of remote sensing have used satellite remote sensing for forest carbon estimation at large scales [13]. However, this approach ignores the extreme environmental heterogeneity of island ecosystems; subtropical islands in particular are often covered in clouds [14], [15], which makes it difficult to capture important cover and structural information [16], [17], leading to insensitivity in national or global models when applied to islands [18], [19]. Consequently, substantial uncertainties remain in island carbon stock assessments derived from both ground and satellite data [20], [21], [22].

The emergence of autonomous aerial vehicle (AAV) remote sensing provides a new technical path for fine-scale forest AGB estimation [23]. AAVs possess core advantages such as low-altitude flight and high maneuverability, enabling them to flexibly overcome the cloud cover constraints of island environments and demonstrating outstanding performance in monitoring surface resources across complex terrains. Kwon et al. [24] accurately inverted cyanobacterial pigment concentrations in reservoirs through AAV-based hyperspectral remote sensing, validating the reliability of AAVs for small-scale ecological monitoring. Qiu et al. [25] used AAVs to establish an integrated airground monitoring network, achieving high-frequency and high-precision monitoring of cyanobacterial blooms in lakes. A primary challenge, however,

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lies in accurately deriving critical AGB parameters, such as tree height and species composition, from AAV data. Although AAV lidar can directly and accurately acquire 3-D structural information [26], its overall cost, including high hardware procurement (typically 5–10 times that of multispectral cameras), complex data processing, and higher technical requirements for operators, poses a challenge for applications with limited budgets or large-scale deployment [27]. In contrast, photogrammetric methods using RGB or multispectral imagery offer a more readily available alternative for generating digital elevation models and canopy height models (CHMs). While these passive optical methods are inherently limited by canopy occlusion [28], their significantly lower cost, ease of deployment, and compatibility with widely available consumer platforms make them a pragmatic and scalable option for mapping forest structure in complex island environments. This occlusion prevents passive sensors from penetrating canopies to accurately model the terrain below, introducing substantial errors (80%–120%) in extracted tree dimensions and subsequent AGB estimates [29], [30]. In investigating the complex topography and patchy distribution of forests on islands, it is of great significance to integrate the vegetation characteristics captured by spectra with the reconstructed 3-D structure of the forest [31], especially under the conditions of different tree species structures [32]. Furthermore, the application of allometric equations requires the establishment of a corresponding intraforest species database [33]. Tree species classification in urban settings can achieve 80%–90% accuracy [34], [35]. However, due to the limited generalization ability of the tree species recognition model, the results of migrating the general large-scale training set established on land to the predicted classification in a specific ecosystem deviate greatly from the actual situation [36]. Hirano et al. [37] used AVIRIS hyperspectral image data to map wetland vegetation species, and the overall map accuracy was only 66%. This defect will be magnified in island ecosystems due to their unique land-sea interactive habitat and environmental isolation [38].

Given these challenges, the overall objective of this study is to use AAVs to collect multispectral images of a typical offshore island, reduce tree height estimation bias through land cover classification, and combine AAV photogrammetry with the Mask R-CNN for tree species identification to establish an innovative AGB estimation workflow for offshore islands. The specific aims are as follows:

1) Characterize the forest structure and species composition of a typical offshore island ecosystem.
2) Assess the propagation of error from AAV data processing methods to overall AGB uncertainty in complex terrain.
3) Evaluate the impact of species-specific allometric equations on AGB estimation results. To our knowledge, this is one of the first studies applying AAV-based deep learning species classification for AGB estimation in offshore island forests. This work provides a critical reference for remote sensing-based carbon accounting of offshore islands and offers insights into the carbon cycling mechanisms of forest ecosystems in land-sea interaction zones.

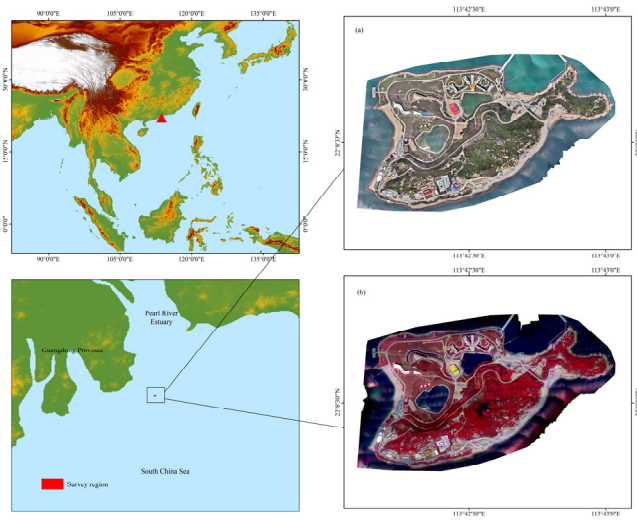


Fig. 1. Approximate location of Triangle Island (marked by a red triangle) within China (highlighted in green). (a) RGB orthomosaic and (b) false-color composite (Nir, Red, Green) orthomosaic acquired from AAV.

II. Materials and Methods

A. Study Area

The study area, Triangle Island (22°08′N, 113°42′E), is located in the northwestern Wanshan Islands in the eastern sea area of Zhuhai City (Fig. 1) and is in the downwind direction of the human-intensive Great Bay Area. Because of its special geographic location, Triangle Island is an offshore island with high-intensity atmospheric deposition transport influence [39]. The study area has a subtropical marine monsoon climate, with an annual rainfall between 1700 and 2200 mm and an average air temperature of about 23.5 °C. Triangle Island covers an area of 0.87 km^2, with a 4.9 km coastline predominantly featuring rocky beaches along its shores. The terrain is rugged, with a flat northwest section formed, low hills dominating the south and east, and the highest elevation (93 m) located centrally on the island's southern side.

The island's diverse vegetation encompasses vine shrubs, vascular plants, and both natural and artificially cultivated trees. Trees on Triangle Island that are not shaded by the upper canopy were selected, including *Casuarina equisetifolia*, *Litsea rotundifolia*, *Avicennia marina*, and *Olea europaea*, which share similarities in morphological characteristics, ecological environment, and growth habits. The island's forests are dominated by low-growing trees, with an average height of less than 4 m, and most trees falling between 2 and 4 m in height.

B. AAV Data Acquisition and Postprocessing

Imagery data acquisition in the study area was performed using a DJI Mavic 3 Multispectral AAV, which met the RGB and multispectral data acquisition requirements of the region (see Supplementary Materials Table SI). The specific time of execution of the aerial photography mission was

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around 1:00 PM on March 26, 2024, during the highest solar altitude, with minimal impact from shadows. The aerial photography was set to have a heading overlap of 70%, a lateral overlap of 60%, a flight altitude of 120 m, and a flight speed of 8 m s^{-1}. AAV RGB and multispectral camera images were constructed using Pix4Dmapper Pro (version 2.0.104). The mean ground resolution of the orthomosaics was 6.44 cm/pix, and the geographic coordinate system used was WGS 84. Band synthesis of a single-band orthomosaic was performed by ArcMap 10.8, saved as a four-band orthomosaic, and vegetation indices were calculated based on Table SII.

Land cover classification was performed using an object-based image analysis approach in eCognition Developer (version 10.3). The multispectral orthomosaic (6.44 cm/pixel) served as the input. The image was first segmented into homogeneous objects using the multiresolution Segmentation algorithm (scale parameter = 10, shape = 0.5, compactness = 0.5), parameters optimized to match canopy structures. Four classes were defined: artificial surface, barren, vegetation, and water. For each object, a suite of 44 potential features was extracted, including spectral mean values from four bands, eight vegetation indices (Table SII), 32 GLCM-based texture measures (Homogeneity, Contrast, etc.), and elevation from the DSM. Feature Space Optimization was used to select the 20 most discriminative features for the final classification, which was executed using a Random Forest classifier (100 trees). We created a validation dataset that does not overlap with the calibration dataset. The reference dataset used for accuracy assessment was established through manual visual interpretation of generated high-resolution AAV orthomosaics. The potential errors and uncertainties of this method mainly stem from two aspects: subjective differences in some land cover transition areas during interpretation, and potential spatial inconsistencies between the instantaneous state represented by the reference data and the classification results. We need to point out that these are inherent constraints in the validation process, but visual interpretation based on ultrahigh-resolution imagery is one of the reliable methods for obtaining validation data in the field of remote sensing. The 138 accuracy assessment points were assigned to their associated zones, and the total number of pixels forming these zones was used as the accuracy assessment dataset for all classifications. The confusion matrix provided information about user and producer accuracy, overall accuracy, and Kappa coefficient. The classification achieved an overall accuracy of 90.58% and a Kappa coefficient of 0.8702 (Table SIII), confirming its suitability for extracting the vegetation mask. The final land cover map was generated in March 2024.

To realize the AGB estimation based on allometric equations, it is necessary to process the 3-D point cloud data derived from AAV images to obtain the alternative parameters of individual tree height and diameter at breast height (DBH). Segmentation of individual trees was used as a basis for extracting individual tree parameters from the AAV remote sensing. Based on the constructed 3-D point cloud model, a DEM, DSM, and CHM were generated by interpolation.

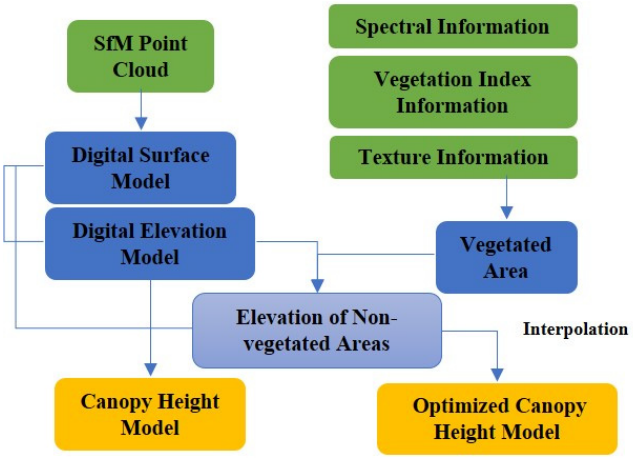


Fig. 2. Optimized CHM generation process conceptual flowchart.

On this basis, the watershed segmentation algorithm was implemented using CHM data to identify the local maximum as the vertex seed of the single tree.

C. CHM Generation and Individual Tree Segmentation

The dense canopy shading leads to missing point clouds on the ground, and DEM generated by direct interpolation has the potential to misclassify vegetation structures as the ground surface. Consequently, there is a systematic bias in CHM (CHM is typically derived by subtracting DEM from DSM), thereby limiting its applicability. As shown in Figs. 2 and S1, we chose to combine land cover classification and interpolation to generate a high-precision DEM. Nonvegetated areas were extracted from the point cloud decoded DEM, and the optimized DEM was derived by interpolating the ground elevation of the vegetated areas based on the adjacent elevation. Additionally, tree heights were extracted from the segmentation of individual trees based on optimized CHM.

The CHM-based watershed segmentation algorithm has been successfully applied to individual tree segmentation in various forest environments. In this study, we employed this method for the automated identification of individual tree crowns within the sample plots. Prior to segmentation, a Gaussian filter was applied to the CHM to smooth minor surface fluctuations and reduce noise. Subsequently, the algorithm identified local maxima on the filtered CHM to serve as the initial seeds for each tree. Following this, the watershed transformation was performed by simulating a water-flow process: starting from these seed points, regions were grown based on the elevation gradient until they reached the boundaries of adjacent catchment basins, thus forming the initial individual tree crown boundaries. To address potential crown overlapping and over-segmentation in complex canopies, a minimum crown area threshold was implemented. Finally, the segmentation results underwent visual inspection and necessary manual refinement to ensure that each segmented polygon corresponded accurately to an individual tree on the ground. This workflow generated precise vector boundaries for all individual trees in the study area, providing the fundamental units for subsequent tree species classification.

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Because of the rugged terrain of Triangle Island and the lack of field measurement data, we conducted internal consistency verification of the photogrammetric point cloud model to evaluate the reliability of the single-tree extraction algorithm. Specifically, we first selected 50 representative trees on the island using a stratified random sampling method. In the high-resolution 3-D visualization environment of the point cloud processing software, we manually identified the ground reference point and the highest crown vertex of each sample tree, and measured the vertical displacement between these two points as the reference tree height. Each tree was measured independently three times, and the average value was used for the final comparison. Then, we compared these manually measured reference tree heights with the tree heights automatically extracted by the single-tree segmentation algorithm through regression analysis.

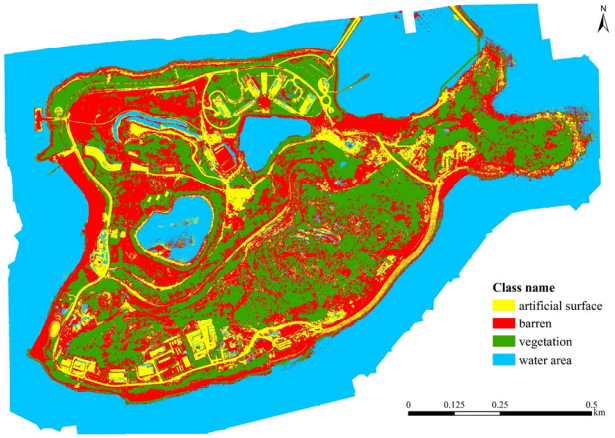


Fig. 3. Spatial distribution of land cover types on Triangle Island.

D. Mask R-CNN-Based Tree Species Identification

Under the complex island patchy background, we employ the Mask R-CNN to identify different canopy features of tree species for tree species identification. This method has excellent detection accuracy and fast segmentation speed, and can detect single tree crowns in complex forest environments, meeting the process requirements of tree species identification in this study. As shown in Fig. S2, the Mask R-CNN has two processing stages: the first stage extracts multiscale feature maps of multispectral images (Green, Red, Red edge, NIR) through the ResNet-50 backbone network; the second stage generates candidate boxes through the region proposal network (RPN), and simultaneously performs object detection, bounding box regression, and pixel-level mask prediction to facilitate tree species identification.

In this study, the Mask R-CNN was constructed based on Detectron2 [40]. First, the "Labelme" image annotation tool was used to manually annotate the four target tree species in the images acquired by the AAV multispectral camera and generate a file in JSON format. Then, all the images were divided into training images (1016) and test images (254). Mask R-CNN was trained using the training dataset, with the number of classes set to five (four tree species + one background image). The model performed 1 20 000 iterations, with a learning rate of 0.001, two images per batch, a ROI header batch size of 1024, and input images uniformly resized to 1024 $\times$ 1024 pixels. By monitoring the loss function during training, we observed that the total loss stabilized after 1 50 000 iterations, indicating that model training was complete and had reached good convergence. A labeled training dataset was used to train the model, including spectral features and location information of different tree species. Based on the trained weights, input multispectral imagery to identify tree species.

E. Forest AGB Estimation Methods

To explore the contribution of disturbances caused by the application of key processing methods for AAV multispectral data in complex terrain to the overall AGB estimation errors of the island, this study constructed a forest AGB estimation method for a typical offshore island.

Applying the allometric equations for four representative tree species to calculate the AGB of the island forests. Considering the composition of the main tree species in the study area, we opted for species-specific allometric equations developed by Tumwebaze et al. [41] employing combined equations incorporating either DBH or tree height measurements (Table I). Correspondingly, DBH cannot be derived directly from 3-D point cloud data. Therefore, we chose the equation of the tree height-DBH relationship developed by González-Jaramillo et al. [28] to displace the calculation of DBH.

The allometric equations are highly species-specific and may cause considerable AGB estimation errors at island scales. We have integrated the Mask R-CNN deep-learning algorithm to facilitate the identification and extraction of dominant tree species within complex forest stand backgrounds (Fig. S3), thereby enabling the development of AGB estimation models that account for species-specific variations.

The aboveground carbon stock was subsequently derived by applying a conversion factor of 0.5 to the estimated AGB, following the IPCC (2006) default value for forest biomass and consistent with previous ecological assessments [42], [43]. This approach ensures that the biomass calculated for the identified tree species is translated into comparable carbon metrics at the ecosystem level.

III. Results

A. Land Cover and Species Classification Results

In this study, the results of land cover categorization of Triangle Island were obtained, as in Fig. 3, showing the different types of land cover on the island. The land cover types were classified into four categories: artificial surfaces (yellow), barren (red), vegetation (green), and water area (blue). Vegetation is widely distributed, accounting for 36.09%. Table SIII shows the confusion matrix and accuracy assessment of the vegetation interpretation of Triangle Island; the overall accuracy of classification is 90.58%, the kappa coefficient is 0.8702, and the accuracy of producer accuracy and user

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TABLE I
SPECIES-SPECIFIC ALLOMETRIC EQUATIONS FOR THE FOUR MAIN TREE SPECIES AND TREE HEIGHT-DBH RELATIONSHIP EQUATION ON THE ISLAND

	Equation	References	Regions	R^2
Species				
Casuarina equisetifolia	$AGB=e^{-2.99-0.79\ln(DBH)+3.27\ln(H)}$	[41]	Tropical rainforest	27.5 %
Olea europaea	$AGB=0.623\times DBH^{1.352}\times H^{0.703}$	[42]	Subtropical evergreen broad-leaved forest	95.4 %
Avicennia marina	$AGB=0.3404\times DBH^{2.0273}$	[43]	Mangrove ecosystem	87.0 %
Litsea rotundifolia	$AGB=e^{-1.472+1.818\ln(DBH)+0.732\ln(H)}$	[44]	Subtropical evergreen broad-leaved forest	93.6 %
H-DBH				
	$DBH=e^{\frac{\ln(H)-0.85}{0.56}}$	[29]		

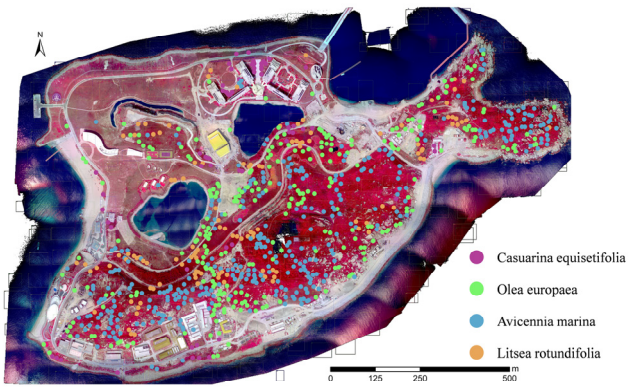


Fig. 4. Employing the Mask R-CNN to identify the spatial distribution of tree species and false-color composite (Nir, Red, Green) orthomosaic. The polygonal box is the output box for tree species identification.

accuracy of each category is more than 80%. The classification map effectively extracts vegetation from the complex island cover, and the classification accuracy meets the needs of subsequent processing of vegetation areas.

Tree species were identified by the Mask R-CNN, and Triangle Island was divided into four categories of tree species detected (Fig. 4) by the optimized CHM within the vegetation area, which were in descending order according to the proportion of tree species: *Avicennia marina* (44.43%), *Olea europaea* (33.04%), *Litsea rotundifolia* (14.73%), and *Casuarina equisetifolia* (7.80%). The tree species composition results revealed the dominant species composition of the island's forest communities, providing key tree species input for subsequent biomass estimation.

B. Individual Tree Segmentation and Structural Parameter Extraction

In this study, the original CHM obtained by AAV photogrammetry and the optimized CHM improved by land cover classification were used to perform single tree segmentation statistics using the watershed algorithm. The optimized CHM yielded a 4.1% higher number of detected trees (1380) compared to the original CHM (1326) (Table II). The extracted mean DBH and tree height variability reflect the unique canopy

TABLE II
CALCULATE THE NUMBER OF INDIVIDUAL TREES, MEAN TREE HEIGHT, AND MEAN DBH BASED ON THE ORIGINAL CHM AND OPTIMIZED CHM

	Individual trees	Mean tree height (m)	Mean DBH (cm)
Original CHM	1326	3.60 ± 1.70	2.49 ± 2.37
Optimized CHM	1380	3.80 ± 1.80	2.74 ± 2.56

structure of the island forest shaped by the environment of the study area. The area is dominated by naturally regenerated low coastal forest and shrubland, which contains many saplings or dwarfs that have not grown well due to the harsh island environment. The high relative variability stems from two aspects. One is the actual mixing of tree sizes in this heterogeneous ecosystem. The second aspect is the methodological limitation inherent to visible-light photogrammetry: its inability to adequately penetrate and capture understory canopies, which leads to unavoidable errors in height extraction for shaded or lower-layer trees. Consequently, DBH is derived indirectly from these potentially biased tree heights through the allometric growth equation [28], a process that amplifies any inaccuracies originating from the CHM, especially in areas with understory vegetation or canopy overlap. Although using the original CHM for single tree segmentation can accurately identify the number of trees at the island scale, the extracted tree height has a relative deviation of −5.6% compared with the reference dataset obtained by using the optimized CHM for single tree segmentation. This height deviation will propagate to the DBH estimation through the allometric growth equation, resulting in a relative deviation of −10.0% in the DBH estimation value. This deviation is not negligible for the statistics of stand parameters at the island scale and will further propagate to the carbon estimation at the ecosystem level [44].

It should be noted that these percentages represent the deviation relative to the reference dataset. The optimized CHM is not presented as an absolute ground truth. Instead, it is employed as a best-available internal reference within the photogrammetric framework. Its primary function is to provide a consistent and methodologically improved baseline against which the original CHM can be evaluated. The reported relative deviations should be interpreted as indicators of the

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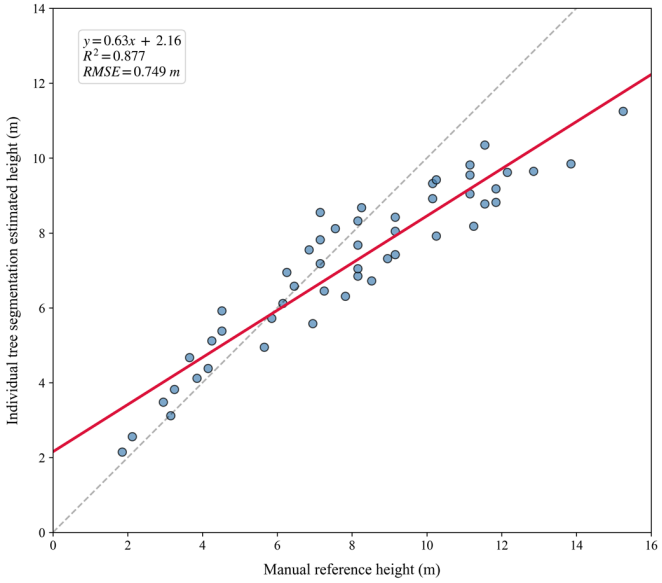


Fig. 5. Regression analysis between manual reference height and individual tree segmentation estimated height.

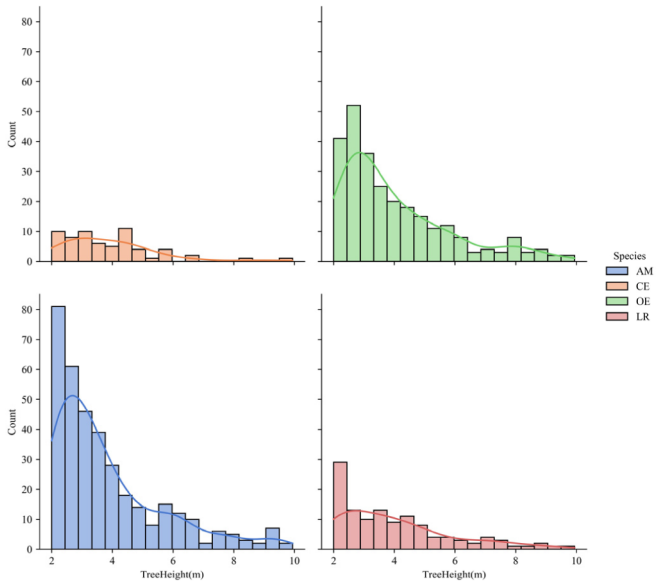


Fig. 6. Frequency distribution of tree heights for four species derived from optimized CHM. AM: Avicennia marina, CE: Casuarina equisetifolia, OE: Olea europaea, LR: Litsea rotundifolia.

improvement achieved by our optimization workflow, not as absolute accuracy measures. This approach aligns with common practice in remote sensing studies where independent field measurements are unavailable, and methodological comparisons rely on internally consistent benchmarks.

To further substantiate the robustness of this internal benchmark, we cross-validated the algorithmic extractions against manual point-cloud measurements for 50 representative sample trees across the island. The results (Fig. 5) revealed a high degree of statistical consistency, yielding a coefficient of determination of 0.88 and a linear regression model of y = 0.63x +2.16. This strong correlation confirms that the optimized algorithm accurately captures the vertical structure of the canopy as recorded in the high-density photogrammetric data. By demonstrating that the algorithmic outputs are highly faithful to the original 3-D geometric information, this verification provides the empirical evidence necessary to justify the optimized CHM as a reliable reference for evaluating the relative deviations (5.6% for tree height and 10.0% for DBH) observed in the original framework.

The tree height frequency distribution relationship (Fig. 6) for each species of tree height parameter extracted by the optimized CHM shows that *Avicennia marina* and *Litsea rotundifolia* are mostly low trees, and *Olea europaea* species as a whole are slightly taller, of which *Casuarina equisetifolia* is in the 2–3 m. *Olea europaea* had the highest mean tree height (4.02 ± 1.80 m), followed by *Litsea rotundifolia* (4.00 ± 1.83 m) and *Casuarina equisetifolia* (3.87 ± 1.47 m). *Avicennia marina* had the highest number of plants identified but also the lowest mean tree height extracted, about 3.82 ± 1.80 m. The four tree species on the island were still dominated by low-growing plants overall, with only sporadic distribution of taller individuals (7–10 m).

TABLE III
AGB ESTIMATION RESULTS USING FOUR SPECIES-SPECIFIC ALLOMETRIC EQUATIONS FOR THE ORIGINAL CHM SOLUTION

Species	Estimation of original results from single tree segmentation		Estimation of vegetation area results for single tree segmentation only	
	Mean AGB per tree (t)	Total AGB (t)	Mean AGB per tree (t)	Total AGB (t)
Casuarina equisetifolia	2.12 ± 2.12	2812.45	2.16 ± 2.02	1521.64
Olea europaea	8.51 ± 16.08	11281.08	8.43 ± 14.71	5940.17
Avicennia marina	4.22 ± 9.46	5593.25	4.10 ± 8.52	2895.41
Litsea rotundifolia	7.73 ± 19.22	10253.88	7.41 ± 17.12	5223.45

Note: Total carbon stock for each species can be derived by applying the conversion factor of 0.5 to the reported AGB.

C. AGB Results Estimated by Different Methods

According to the original CHM which estimated the AGB of each of the four allometric equations as a uniform tree species across the island as in Table III, there was an overestimation of 45.90%–49.06% of the estimated AGB from the original results of the segmentation individual trees compared to the estimated AGB from the results of the segmentation individual trees for the statistics of only the vegetation areas on a whole island scale. On the other hand, from the point of view of single tree average statistics, there were both overestimates and underestimates, with the most significant occurrence of overestimation for *Litsea rotundifolia*, which amounted to 4.19%, and underestimation for *Casuarina equisetifolia*, which existed at 1.74%.

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TABLE IV
AGB ESTIMATION RESULTS USING FOUR SPECIES-SPECIFIC ALLOMETRIC EQUATIONS FOR THE OPTIMIZED CHM SOLUTION

Species	Estimation of original results from single tree segmentation		Estimation of vegetation area results for single tree segmentation only	
	Mean AGB per tree (t)	Total AGB (t)	Mean AGB per tree (t)	Total AGB (t)
Casuarina equisetifolia	2.35 ± 2.30	3241.34	2.45 ± 2.25	1982.04
Olea europaea	9.98 ± 17.83	13776.72	10.36 ± 17.13	8372.34
Avicennia marina	5.04 ± 10.57	6951.00	5.18 ± 10.07	4184.20
Litsea rotundifolia	9.33 ± 21.58	12879.14	9.52 ± 20.42	7688.96

Note: Total carbon stock for each species can be derived by applying the conversion factor of 0.5 to the reported AGB.

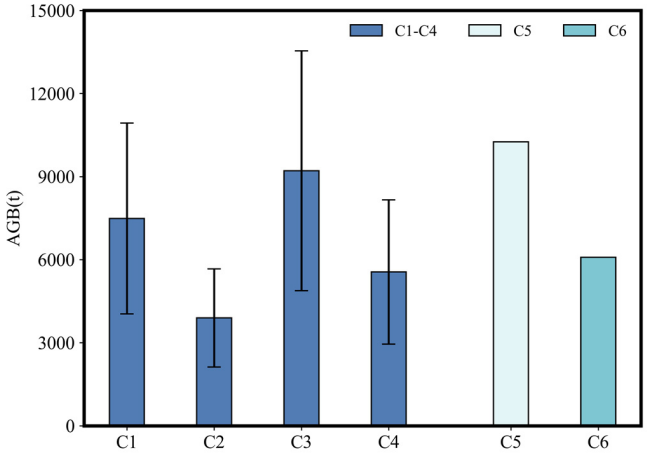


Fig. 7. Differences in AGB calculation using different methods. C1, AGB estimation using original CHM individual tree segmentation; C2, AGB estimation using original CHM individual tree segmentation with vegetation area screening; C3, AGB estimation using optimized CHM based individual tree segmentation; C4, AGB estimation using optimized CHM-based individual tree segmentation with vegetation area screening; C5, AGB estimation using optimized CHM-based individual tree segmentation combined with tree species identification; C6, AGB estimation using optimized CHM-based individual tree segmentation with vegetation area screening and tree species identification.

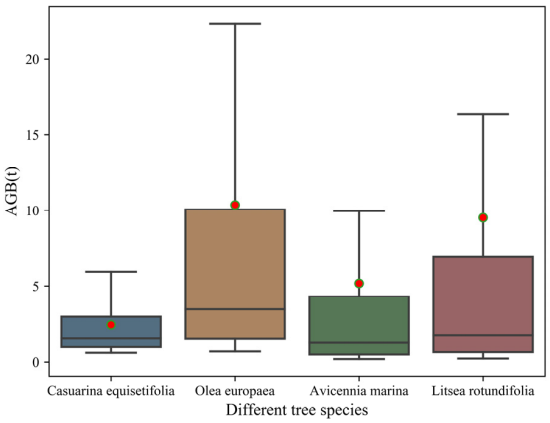


Fig. 8. Individual tree AGB estimated by the optimized method. AGB estimation using optimized CHM-based individual tree segmentation with vegetation area screening and tree species identification.

Comparing the results of the four allometric equations in the extrapolation of tree height and DBH in the AGB, the gap between the estimated AGB results can reach almost three times. The species with the highest imputed AGB was *Olea europaea*, followed by *Litsea rotundifolia*, *Avicennia marina*, and *Casuarina equisetifolia*.

Comparing the AGB estimation results of the original CHM and the optimized CHM using allometric equations of the four tree species (Table IV), it can be seen that the original CHM has a great underestimation of the AGB, which will be magnified in the AGB estimation of the vegetated areas. For the results of the whole island, it seems that the original CHM will underestimate the AGB by nearly 13.23%–20.38%, and for the results of only counting the vegetated areas, the original CHM will underestimate the AGB by nearly 23.23%–32.07%.

Fig. 7 demonstrates the differences between the different methods of calculating AGB, C1, C2, C3, and C4, which were divided into four categories of tree species based on the treatments of tree species identification for statistical purposes, which were further divided into two categories of original CHM and optimized CHM. Analysis showed that optimized

CHM generally improved AGB estimates. For example, in C1 and C3, OE increased from 11 281.1 t to 13 776.7 t, AM increased from 5593.2 t to 6951.0 t, LR increased from 10 253.9 t to 12 879.1 t, and CE increased from 2812.5 t to 3241.3 t, indicating that the optimized model more accurately captures canopy height and distribution. Vegetation area screening reduced estimates. For example, compared to C1, OE decreased by 5340.9 t, AM decreased by 48.2%, LR decreased by 49.1%, and CE decreased by nearly 50%. Compared to C3, AGB values for all tree species decreased by approximately 40% in C4, as screening for vegetation cover identified excluded individual trees that were incorrectly labeled in nontarget vegetation areas. C5 was AGB estimation using optimized CHM-based individual tree segmentation combined with tree species identification, and C6 was AGB estimation using optimized CHM-based individual tree segmentation with vegetation area screening and tree species identification; the results were 10 258.42 and 6088.43 t, respectively.

They introduced tree species identification and quantified the impact of species-specific allometric equations on AGB estimation. The integrated approach (C6: Optimized CHM + Mask R-CNN species identification) provided the most refined AGB estimates. Because the interference of irrelevant nontarget tree species or vegetation on the estimation is eliminated, the influence of the allometric growth equation established for a single forest community is reduced in terms of estimation precision. *Olea europaea* contributed the most to the total AGB (48.30%–53.65%), followed by *Avicennia marina* (26.76%–29.53%) and *Litsea rotundifolia* (17.04%–19.80%).

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The estimated AGB for trees in vegetated areas after species identification was selected as a reference can be seen that using only four very different allometric equations as a uniform tree species across the island would introduce perturbations, and that the allometric equations of *Casuarina equisetifolia* and *Avicennia marina* would contribute to the island-scale AGB estimation by about −68% and −32% underestimation, while manifestation in *Olea europaea* and *Litsea rotundifolia* results in approximately +36% and +26% overestimation.

To further clarify the sources of differences between different estimation methods, we calculated the differences in AGB based on different tree species. As shown in Fig. 8, for the AGB of individual trees estimated from different tree species, the average AGB of individual trees decreased from 10.4 to 2.5 t as the species changed. AGB of individual *Casuarina equisetifolia* ranged from 0.6 to 6.0 t; the AGB of individual *Olea europaea* ranged from 0.7 to 22.4 t. Individual *Avicennia marina* ranged from 0.2 to 10.0 t; the AGB of the single *Litsea rotundifolia* was 0.2–16.4 t, forming a unique biomass allocation pattern due to the high salinity environment and flooding conditions. From the maximum and minimum values and the distribution, the range of fluctuation of the AGB is more pronounced with the change of tree species. The estimated fluctuations of AGB at the single tree scale ranged from 3.8 to 8.5 t. The range of these values reflects the carbon storage potential of mangroves and different evergreen tree species in island terrains, which means that a single empirical approach that ignores tree species is a significant source of uncertainty in AGB estimates.

IV. Discussion

A. Species Composition and Individual Tree Parameters of Triangle Island Forest

This study used AAV multispectral data for photogrammetry, extracted and solved single tree parameters from the derived CHM, and used deep-learning models for tree species identification based on spectral information. Species composition determines the proportion of different tree species in a forest. Different tree species have varying growth characteristics and biomass accumulation capacities, directly impacting overall AGB. Three-dimensional structural characteristics of trees, such as tree height and DBH, are commonly used as key input variables for normalization in AGB estimation studies. These parameters are correlated with biomass. Obtaining continuous and accurate data on tree species composition and 3-D structure in a replicable and flexible manner will facilitate future coastal and inland forest monitoring.

The forest of the Triangle Island is mainly composed of small trees, as shown in Fig. S4, which is consistent with the findings of Mohammadi et al. [45], that natural vegetation presents an inverse J-shaped population structure as the population expands or develops. Among the species identified, *Avicennia marina* and *Olea europaea* account for about three-quarters of the tree species (44.43% and 33.04%), which are the most dominant tree species on Triangle Island. It can be considered that the vegetation morphology and phenology of the above two tree species indirectly control the ecological

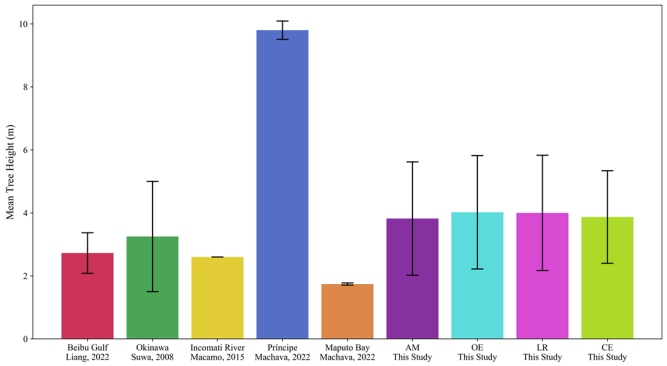


Fig. 9. Average tree height of the forest in different islands.

environment of the entire island. However, existing studies have pointed out that in tropical and subtropical regions, species composition and abundance should be diverse [46], while inland forests, due to their wide distribution and diverse climate types, have a richer tree species composition [47]. The method proposed in this study for drawing forest maps at the species level requires detailed forest surveys. Because of this limitation, only the four most representative tree species were selected for characterization in this study. However, in the study of small and medium-sized forestry, we still believe that drone tree species identification and photogrammetry are promising research directions for remote sensing monitoring of coastal and inland forests.

As described in Section II-A, the forest on Triangle Island is mainly composed of low trees with an average height of less than 4 m. This is consistent with the average plant height of forest communities in coastal areas of islands reported in the Beibu Gulf and Okinawa studies [14], [15], [16], as shown in Fig. 9. A possible explanation is that the harshness of the special habitats on the islands significantly inhibits the productivity of trees, resulting in dwarfing characteristics of the forest stands [19]. It is worth noting that the tree height difference between Principe Island and Maputo Bay is that the forests on Principe Island are mainly composed of *Rhizophora harrisonii*, while the forests in Maputo Bay are mainly composed of *Avicennia marina*, which is more similar to the main tree species structure of Triangle Island in this study, so their tree height is closer to the estimated value of this study [32].

B. AAV Multispectral as an Effective Method for Estimating AGB of Offshore Island Forests

This study attempts to estimate the AGB of offshore island forests using deep learning tree species identification and forest structural parameter extraction. Tree species recognition or photogrammetry of forests is a relatively new technique that has received increasing attention in recent years, especially as a tool for measuring carbon sinks in forest sample plots. Table V comprehensively compares the technical characteristics and application scenarios of RGB, multispectral, and LiDAR in forest AGB estimation, revealing the advantages and disadvantages of different sensors in forest inventory

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TABLE V
COMMON APPLICATIONS OF AGB OBSERVATION ESTIMATION METHODS

Sensor	Description	Common applications in AGB estimation	References
RGB	High resolution and low cost, includes three visible bands (Red, Green, and Blue).	Identify and delineate vegetation boundaries.	[48]
		Calculate vegetation indices based on Red, Green, and Blue.	[49]
		Create a DEM.	[50]
		Create a CHM and determine canopy structure.	[51]
Multispectral	Provides additional information on the Red edge and NIR bands compared to RGB.	Calculate various vegetation indices, such as NDVI.	[52]
		Create a DEM.	[53]
		Create a CHM and ascertain canopy structure.	[54]
LiDAR	Emits laser pulses and measures their reflection time to obtain 3D spatial data.	Create a more accurate DEM.	[55]
		Create a more accurate CHM to calculate canopy structure.	[56]

and their applicable boundaries. RGB technology is widely used in vegetation boundary extraction [48] and canopy structure modeling [51] due to its high resolution and low cost. However, its reliance on RGB bands (Red, Green, and Blue) leads to two defects: limited vegetation index construction and insufficient terrain correction capabilities [49], [50]. In this study, multispectral technology has performed well in three aspects: cover classification (Fig. 3), use of optimized CHM (Tables III and IV), and screening of single trees in nonvegetated areas (Fig. 8). Although multispectral canopy penetration in dense forests is weaker than LiDAR, its comprehensive cost-effectiveness and ease of use make it the first choice at the island scale [56]. However, due to differences in remote sensing equipment and postprocessing methods of remote sensing data, the variability of estimated values between different methods is unclear, which limits our accurate assessment of forest AGB [57]. This variability can be quantitatively analyzed through two key dimensions, namely allometric equations and areal statistics.

This study estimates the AGB of typical offshore islands using different allometric equations and areal statistics methods, hoping to analyze the usability of carbon stock accounting for offshore islands using AAVs. General allometric equations may be susceptible to strong and uncontrolled local biases [58]. In this study, AAVs were used to map established populations and comparative experiments with species-specific allometric equations were performed to quantify errors and offset their propagation.

At the tree level, the study by Mulatu et al. [38] overestimated AGB by 9.6%–77.8% using pantropical allometric equations, while the species-specific allometric equations had an estimated error of only 0.1%–7.9% for each tree species, illustrating the potential improvement offered by species-specific approaches. In this study, employing species-specific equations for the wrong tree species resulted in an estimated deviation ranging from −68% to +36% (Fig. 7) from the values derived via our optimized method. This extreme range underscores that the use of the allometric equations is the main source of uncertainty in the AGB estimates for trees and plots. The findings highlight a general caution: uncritical application of generalized allometric equations, especially in unique ecosystems, can lead to serious biases in carbon accounting. The allometric equations using a nonvariable DBH

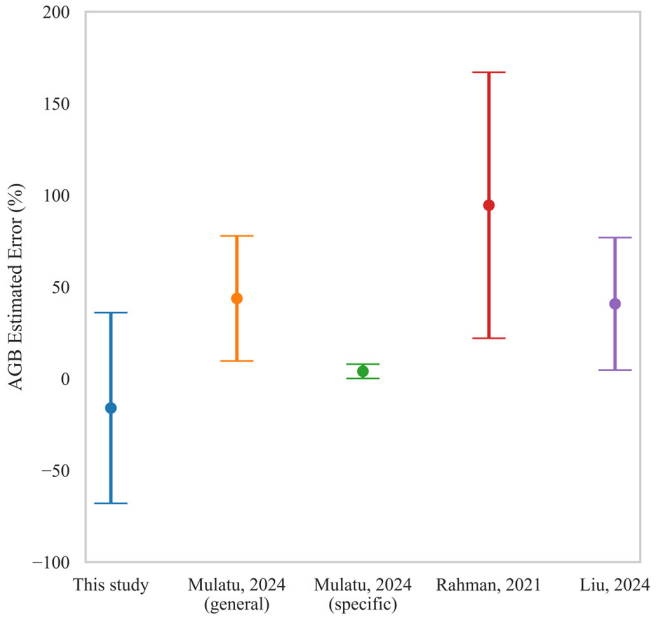


Fig. 10. Error range of AGB estimation in this study and published studies [17], [20], [38].

to tree height have been shown to lead to an overestimation of total AGB by 10%–20% [59]. Mitchard et al. [19] pointed out that the magnitude of the allometric equations bias will have a spatial pattern, depending on the forest type and the regional differences in forest structure and allometric equations. AGB estimates carried out in different regions will also accumulate the uncertainty of the spatial pattern on the original uncertainty of the allometric equations. As shown in Fig. 10, in the study of Rahman et al. [17], the general allometric equations also overestimated AGB by 22%–167% compared with the species-specific allometric equations.

One aspect worth noting is that, unlike similar studies that usually report an overestimation of the error range for AGB estimates [20], this study showed a significant underestimation in AGB estimates. This study used segmentation of individual trees at the tree level within the island, and there were a few incomplete segmentations that led to incomplete tree inventories. In particular, offshore islands are often dotted with

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tilted tree trunks that span multiple vertical columns. Our method may fail to detect tree trunks with large tilt angles, leading to an underestimation of AGB. This can be partially avoided by integrating field validation points.

The results of C2, C4, and C6 (Fig. 7) show that by screening the vegetated areas, the data interference from bare ground or nonvegetated areas was excluded, which effectively avoided the problem of high AGB estimation due to the erroneous participation of nonvegetated data. However, this study did not participate in destructive sample statistics, which was limited by the protection policy of the study area. Furthermore, the species classification model is limited to the species it was trained on and may not generalize well to new locations with different species without additional training data. Overall, it is crucial to choose appropriate allometric equations in carbon storage accounting for offshore island forest ecosystems.

In addition, due to the huge contribution of forest structure indicators to forest AGB, which is reflected in the estimated fluctuation of AGB from 13.23% to 32.07% in this study, it is extremely important to detect the accurate location of the forest CHM and subcanopy terrain DEM [60]. This study uses a method to optimize DEM to obtain a more realistic CHM. The two key points of this method are multispectral vegetation area classification extraction and ground elevation interpolation of vegetation areas. The reliability of vegetation structure indicators depends largely on the accuracy of identifying vegetation area boundaries [61]. The main disadvantage lies in the inherent limitations of photogrammetry. The accuracy of the CHM is contingent on the ability to model the ground surface, which is obstructed by vegetation. While our optimization method mitigates this, it cannot fully overcome the problem in very dense forests, where LiDAR's ability to penetrate canopy gaps would be superior [56]. The land cover classification we adopted is object-oriented classification, which is suitable for high-resolution images such as drone images and has an accuracy of 93.75% for vegetation classification and recognition on islands. This improved method is suitable for CHM mapping of vegetation communities in island areas, because these areas usually have low and dense canopies, and the ground surface can be effectively identified from orthomosaic to obtain ground elevation [62]. The study by Fekry et al. [63] was conducted in an artificial forest with a homogeneous forest structure, where multisource data fusion reduced the rRMSE of tree height estimation by 9.01%. In contrast, natural forests retain complex plots with mixed species and various tree patterns and attributes. which can make full use of the coverage of multispectral information to accurately land cover classification on a 2-D plane for CHM interpolation [64]. For different contexts, such as extensive mainland forests, the method's reliance on high-resolution AAV data may present scalability constraints. However, using object-based image analysis to refine DEM and deep learning for species classification could be adapted to airborne or even very-high-resolution satellite platforms for larger-scale applications.

One limitation of this study is the lack of independent, field-measured tree height data for absolute validation. The accuracy of tree height extraction depends on the quality of the CHM and its segmentation parameters, both of which are affected by canopy shading and inherent limitations of photogrammetry. Future research should prioritize collecting a small amount of field-measured tree height data from easily accessible plots for validation. Alternatively, integrating AAV LiDAR data, which can penetrate the canopy to obtain more accurate topographic and tree height information, could allow for a more rigorous evaluation of photogrammetric methods.

This research extends beyond the specific case of the Triangle Island. The proposed method, which integrates AAV multispectral photogrammetry, optimized CHM generation, and deep learning-based tree species identification, provides a scalable framework for high-resolution carbon storage assessment in similar environments, including other nearshore islands, coastal forests, or highly heterogeneous fragmented landscapes. The ability to generate accurate species-level AGB maps without relying on destructive sampling is particularly important in protected areas or inaccessible regions.

For broader application, three key prerequisites are necessary. First, access to high-resolution AAV multispectral or RGB imagery. Second, a local or regional tree species inventory or spectral library for training the species taxonomy model. Finally, basic ground reference data for validating land cover and tree height extraction. While this method is optimized for low-lying, dense canopy islands, it can also be applied to continental regions. With the availability of high-resolution topographic and canopy models, it can be used for forest research. Future integration with AAV LiDAR is expected to further enhance the model's robustness across different forest structures.

C. Comparison of AGB Estimation With Other Forest Types

This article considers that the average AGB of Triangle Island is 391.29 t ha^{-1}. The implications of this research extend beyond the specific case of Triangle Island. For similar contexts, such as other offshore islands, coastal forests, or fragmented landscapes with high heterogeneity, our methodology provides a scalable and cost-effective framework for high-resolution carbon stock assessment. The ability to generate accurate, species-level AGB maps without reliance on destructive sampling is particularly valuable in protected or inaccessible areas. Comparing the AGB estimated in this study with various forests around the world (Table VI), we found that the AGB of Triangle Island is within the AGB range estimated for other common coastal forests or inland forests (73–446 t ha^{-1}). AGB of Triangle Island is close to that of the Himalayan temperate forest (369 t ha^{-1}), reflecting similar levels of AGB estimation under moderate disturbance or natural succession conditions [65]. Moderate tree felling can provide growth space for other plants, promote the growth of a variety of plants, and thus maintain regional AGB at a high level [72]. The AGB of the artificial forests in Sehanba (186 t ha^{-1}) and Ximen Island (182 t ha^{-1}) were significantly lower than those in this study. The main reason is that the tree species structure of the artificial forests is single, and the biological structure determines that the carbon sequestration capacity is not as good as that of natural tree species. On the other hand, the forest age is short, and the trees have not yet entered the peak period of AGB estimation. Generally,

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TABLE VI
COMPARISON OF ESTIMATED AGB (T HA^{-1}) OF FORESTS AROUND THE WORLD

Region	AGB (t ha^{-1})	References
Triangle Island	391	This study
Planted forest in the Sehanba forest	186	[66]
The central Terai region of Nepal	446	[67]
Himalayan temperate forest	369	[68]
Mediterranean riparian forest (Acacia dealbata)	251	[69]
Mediterranean riparian forest (Alnus glutinosa)	162	[69]
Mediterranean riparian forest (Salix salviifolia)	73	[69]
Ximen Island	182	[70]
Forests in the contiguous 48 United States	125	[71]

forest AGB will show exponential growth with the increase in tree diameter, but with the increase in tree density, the AGB will decrease accordingly [73]. Large trees contribute more to the overall AGB of the forest. The AGB of the central Terai region of Nepal (446 t ha^{-1}) is higher due to its tropical climate, high-productivity tree species, and less human disturbance [67]. AGB is also closely related to environmental factors. The higher precipitation and moderate light radiation in these areas may have created the current biogeographic distribution pattern, which in turn promoted the growth of trees and affected the AGB and carbon storage capacity of the region [74]. Triangle Island is the frontier where freshwater (Pearl River) and seawater (South China Sea) meet. This unique habitat supports a mixed distribution of salt-tolerant mangrove forests (*Avicennia marina*) and terrestrial evergreen broad-leaved forests (*Casuarina equisetifolia*, *Litsea rotundifolia*, and *Olea europaea*). The high carbon burial efficiency of mangroves may significantly increase the AGB of island areas, which means that islands in estuary locations have huge carbon storage potential [75].

There are many possible explanations for the differences between the AGB estimated in these different regions and this study, but we focus on the three explanations that we believe are most likely to dominate. First, forest growth and composition have significant regional differences. Natural forests in different climatic zones will show huge differences in tree spatial distribution and species diversity, which will significantly affect the productivity of forest land and the carbon sequestration capacity of forests [76]. Second, differences in data processing and filtering methods. If the data or datasets are significantly different, they should be filtered. In large-scale AGB estimation, especially when using satellite datasets, the data resolution is very small, and nonforest information is always averaged when processing ground units [77]. This averaging process will reduce noise and eliminate differences in large-scale data processing to a certain extent, but the residual bias of this process may be passed on to maps, resulting in an overestimation of AGB [78]. Finally, use different data for estimation. Kc et al. [67] used field survey data in addition to LiDAR data. The research of Sa et al. [66] was mainly driven by Landsat 8, Sentinel-2A, and ALOS2 PALSAR. Obviously, the AGB values estimated for the same plot using LiDAR and multispectral data will produce different results, and some results are even far apart, which is driven by the characteristics of the plot. Future research should focus on integrating AAV-based lidar with multispectral imagery. This can synergize the vertical mapping capabilities of lidar with the classification advantages of multispectral imagery, potentially leading to more accurate estimates of AGB. This remains a critical challenge requiring further investigation to advance the synergistic integration of multisource remote sensing data for robust large-scale regional monitoring.

V. CONCLUSION

Based on multispectral remote sensing technology from AAVs combined with the Mask R-CNN and allometric equations, this study refined the AGB of the forest in Triangle Island and explored the impact of different processing methods on the final AGB estimation results. Our results show that ignoring the differences in allometric equations of tree species will lead to large errors in AGB estimation (−68% to +36%), which is the main source of uncertainty in AGB estimation for offshore islands. The Mask R-CNN, based on multispectral data, effectively identified four major tree species, of which Avicennia marina and Olea europaea accounted for more than 70% of the total AGB, highlighting the necessity of using species-specific allometric equations in carbon stock accounting. By combining the optimized CHM with the vegetation regional interpolation method, the study further revealed that, compared to the optimized CHM, the original CHM yielded relative deviations of approximately −5.6% in tree height and −10.0% in DBH, highlighting the impact of canopy occlusion on photogrammetric height estimation. These findings provide a more reliable methodological basis for AGB estimation in areas with similar forest structures, such as islands. In the estimation of AGB or carbon storage of island forests, more emphasis should be placed on collecting information on specific dominant species, followed by accurate measurement of DBH or tree height, so that we can estimate AGB within a controllable uncertainty range. In future studies, methods for tree species identification and forest structure parameter extraction should be further studied to develop more accurate indicators for the estimation of forest AGB using AAV multispectral remote sensing technology, which may greatly improve the estimation of terrestrial AGB in local islands, regional ecosystem, or on a global scale.

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